

Лабораторная работа № 12. Настройка NAT

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Информация

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Цель работы

Приобретение практических навыков по настройке доступа локальной сети к внешней сети посредством NAT.

Задание

Задание

1. Сделать первоначальную настройку маршрутизатора provider-gw-1 и коммутатора provider-sw-1 провайдера: задать имя, настроить доступ по паролю и т.п.
2. Настроить интерфейсы маршрутизатора provider-gw-1 и коммутатора provider-sw-1 провайдера.
3. Настроить интерфейсы маршрутизатора сети «Донская» для доступа к сети провайдера.
4. Настроить на маршрутизаторе сети «Донская» NAT с правилами.
5. Настроить доступ из внешней сети в локальную сеть организации.
6. Проверить работоспособность заданных настроек.
7. При выполнении работы необходимо учитывать соглашение об именовании.

Выполнение лабораторной работы

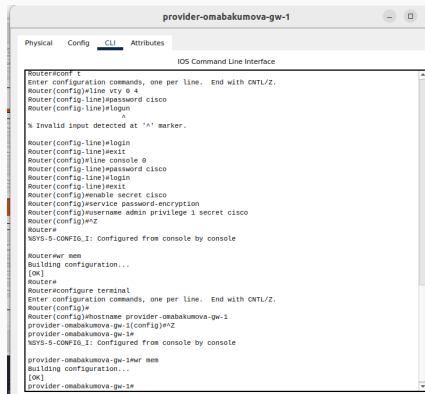
Таблица 1: Распределение внешних ip-адресов

IP-адреса	Примечание	VLAN
198.51.100.0/28	Выделено провайдером	4
198.51.100.1	Маршрутизатор провайдера	
198.51.100.2	msk-donskaya-gw-1	
198.51.100.2–198.51.100.14	Пул адресов для NAT	
198.51.100.2	Web	
198.51.100.3	File	
198.51.100.4	Mail	

Таблица 2: Таблица VLAN

№ VLAN	Имя VLAN	Примечание
1	default	Не используется
2	management	Для управления устройствами
3	servers	Для серверной фермы
4	nat	Линк в Интернет
5-100		Зарезервировано
101	dk	Дисплейные классы (ДК)
102	departments	Кафедры
103	adm	Администрация
104	other	Для других пользователей

Выполнение лабораторной работы



```
provider-omabakumova-gw-1
Physical Config CLI Attributes
IOS Command Line Interface
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#line vty 0 4
Router(config-line)#password cisco
Router(config-line)#login
Router(config-line)#login
Router(config-line)#exit
Router(config)#enable secret cisco
Router(config)#service password-encryption
Router(config)#username admin privilege 1 secret cisco
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr mem
Building configuration...
[OK]
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#hostname provider-omabakumova-gw-1
provider-omabakumova-gw-1(config)#end
provider-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-omabakumova-gw-1#wr mem
Building configuration...
[OK]
provider-omabakumova-gw-1#
```

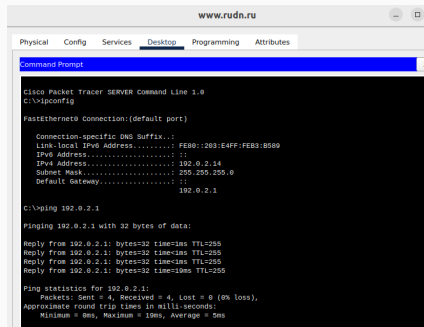
Рис. 1: Первоначальная настройка маршрутизатора provider-omabakumova-gw-1

Выполнение лабораторной работы

```
^C
provider-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
provider-omabakumova-gw-1(config)#interface fa/0
provider-omabakumova-gw-1(config-if)#no shutdown
provider-omabakumova-gw-1(config-if)#exit
provider-omabakumova-gw-1(config)#interface fa/0.4
provider-omabakumova-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.4, changed state to up
provider-omabakumova-gw-1(config-subif)#encapsulation dot1q 4
provider-omabakumova-gw-1(config-subif)#ip address 198.51.100.1 255.255.255.240
provider-omabakumova-gw-1(config-subif)#description mks-donskaya
provider-omabakumova-gw-1(config-subif)#exit
provider-omabakumova-gw-1(config)#interface fa/1
provider-omabakumova-gw-1(config-if)#no shutdown
provider-omabakumova-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
provider-omabakumova-gw-1(config-if)#ip address 192.0.2.1 255.255.255.0
provider-omabakumova-gw-1(config-if)#description internet
provider-omabakumova-gw-1(config-if)#exit
provider-omabakumova-gw-1(config)#exit
provider-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
provider-omabakumova-gw-1#r
Building configuration...
[OK]
provider-omabakumova-gw-1#
```

Рис. 2: Настройка интерфейсов маршрутизатора provider-omabakumova-gw-1

Выполнение лабораторной работы



The screenshot shows a Cisco Packet Tracer interface with a 'Command Prompt' window open. The window title is 'www.rudn.ru'. The tabs at the top are 'Physical', 'Config', 'Services', 'Desktop' (selected), 'Programming', and 'Attributes'. The Command Prompt shows the following output:

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::203:E4FF:FEB3:8589
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.0.2.14
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                192.0.2.1

C:\>ping 192.0.2.1

Pinging 192.0.2.1 with 32 bytes of data:

Reply from 192.0.2.1: bytes=32 time=1ms TTL=255
Reply from 192.0.2.1: bytes=32 time<1ms TTL=255
Reply from 192.0.2.1: bytes=32 time<1ms TTL=255
Reply from 192.0.2.1: bytes=32 time=10ms TTL=255

Ping statistics for 192.0.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 5ms
```

Рис. 3: Пингование provider-omabakumova-gw-1

Выполнение лабораторной работы

```
provider-omabakumova-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
provider-omabakumova-sw-1(config)#line vty 0 4
provider-omabakumova-sw-1(config-line)#password cisco
provider-omabakumova-sw-1(config-line)#login
provider-omabakumova-sw-1(config-line)#exit
provider-omabakumova-sw-1(config)#line console 0
provider-omabakumova-sw-1(config-line)#password cisco
provider-omabakumova-sw-1(config-line)#login
provider-omabakumova-sw-1(config-line)#exit
provider-omabakumova-sw-1(config)#enable secret cisco
provider-omabakumova-sw-1(config)#service password-encryption
provider-omabakumova-sw-1(config)#username admin privilege 1 secret cisco
provider-omabakumova-sw-1(config)#^Z
provider-omabakumova-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-omabakumova-sw-1#r mem
Building configuration...
[OK]
provider-omabakumova-sw-1#
```

Рис. 4: Первоначальная настройка коммутатора provider-omabakumova-sw-1

Выполнение лабораторной работы

```
provider-omabakumova-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
provider-omabakumova-sw-1(config)#interface f0/1
provider-omabakumova-sw-1(config-if)#switchport mode trunk

provider-omabakumova-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

provider-omabakumova-sw-1(config-if)#exit
provider-omabakumova-sw-1(config)#interface f0/2
provider-omabakumova-sw-1(config-if)#switchport mode trunk

provider-omabakumova-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

provider-omabakumova-sw-1(config-if)#exit
provider-omabakumova-sw-1(config)#vlan 4
provider-omabakumova-sw-1(config-vlan)#name nat
provider-omabakumova-sw-1(config-vlan)#exit
provider-omabakumova-sw-1(config)#interface vlan4
provider-omabakumova-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan4, changed state to up

provider-omabakumova-sw-1(config-if)#no shutdown
provider-omabakumova-sw-1(config-if)#exit
provider-omabakumova-sw-1(config)#^Z
provider-omabakumova-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-omabakumova-sw-1#wr mem
```

Рис. 5: Настройка интерфейсов коммутатора provider-omabakumova-sw-1

```
!
interface FastEthernet0/1
 switchport mode trunk
!
interface FastEthernet0/2
 switchport mode trunk
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
```

Рис. 6: Проверка настройки интерфейсов коммутатора provider-omabakumova-sw-1

Выполнение лабораторной работы

```
msk-donskaya-omabakumova-gw-1>en
Password:
msk-donskaya-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-gw-1(config)#interface f0/1
msk-donskaya-omabakumova-gw-1(config-if)#no shutdown

msk-donskaya-omabakumova-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

msk-donskaya-omabakumova-gw-1(config-if)#exit
msk-donskaya-omabakumova-gw-1(config)#interface f0 /1.4
                                     ^
% Invalid input detected at '^' marker.

msk-donskaya-omabakumova-gw-1(config)#interface f0/1.4
msk-donskaya-omabakumova-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.4, changed state to up

msk-donskaya-omabakumova-gw-1(config-subif)#encapsulation dot1Q 4
msk-donskaya-omabakumova-gw-1(config-subif)#ip address 198.51.100.2 255.255.255.240
msk-donskaya-omabakumova-gw-1(config-subif)#description Internet
msk-donskaya-omabakumova-gw-1(config-subif)#exit
msk-donskaya-omabakumova-gw-1(config)#exit
msk-donskaya-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-gw-1(config)#ip route 0.0.0.0 0.0.0.0 198.51.100.1
msk-donskaya-omabakumova-gw-1(config)#exit
msk-donskaya-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-omabakumova-gw-1#
```

Рис. 7: Настройка интерфейсов маршрутизатора msk-donskaya-omabakumova-gw-1

Выполнение лабораторной работы

```
!
interface FastEthernet0/0.104
description other
encapsulation dot1Q 104
ip address 10.128.0.1 255.255.255.0
ip access-group other-in in
!
interface FastEthernet0/1
no ip address
duplex auto
speed auto
!
interface FastEthernet0/1.4
description internet
encapsulation dot1Q 4
ip address 198.51.100.2 255.255.255.240
!
interface Vlan1
no ip address
shutdown
!
router rip
!
ip classless
ip route 0.0.0.0 0.0.0.0 198.51.100.1
```

Рис. 8: Проверка настройки интерфейсов на msk-donskaya-omabakumova-gw-1

Выполнение лабораторной работы

```
msk-donskaya-omabakumova-gw-1>en
Password:
msk-donskaya-omabakumova-gw-1#ping 198.51.100.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 198.51.100.1, timeout is 2 seconds:
..!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 0/0/0 ms

msk-donskaya-omabakumova-gw-1#ping 198.51.100.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 198.51.100.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/4/21 ms

msk-donskaya-omabakumova-gw-1#
```

Рис. 9: Пингование маршрутизатора провайдера

Выполнение лабораторной работы

```
msk-donskaya-omabakumova-gw-1#en
Password:
msk-donskaya-omabakumova-gw-1#ping 100.51.100.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.51.100.1, timeout is 2 seconds:
..!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 0/0/0 ms

msk-donskaya-omabakumova-gw-1#ping 100.51.100.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.51.100.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/4/21 ms

msk-donskaya-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-gw-1(config)#ip nat pool main-pool 100.51.100.2 100.51.100.14 netmask 255.255.255.240
msk-donskaya-omabakumova-gw-1(config)#ip access-list extended nat-inet
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#remark dk
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#permit tcp 10.128.3.0 0.0.0.255 host 192.0.2.11 eq 80
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#permit tcp 10.128.3.0 0.0.0.255 host 192.0.2.12 eq 80
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#remark departments
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#permit tcp 10.128.4.0 0.0.0.255 host 192.0.2.13 eq 80
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#remark adm
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#permit tcp 10.128.5.0 0.0.0.255 host
% Incomplete command.
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#permit tcp 10.128.5.0 0.0.0.255 host 192.0.2.14 eq 80
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#remark admin
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#permit ip host 10.128.6.200 any
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#Z
msk-donskaya-omabakumova-gw-1#
NDVS-5-COMP10.1: Configured from console by console

msk-donskaya-omabakumova-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-omabakumova-gw-1#
```

Рис. 10: Настройка на маршрутизаторе сети «Донская» NAT с правилами

Выполнение лабораторной работы

```
ip access-list extended nat-inet
remark dk
permit tcp 10.128.3.0 0.0.0.255 host 192.0.2.11 eq www
permit tcp 10.128.3.0 0.0.0.255 host 192.0.2.12 eq www
remark departments
permit tcp 10.128.4.0 0.0.0.255 host 192.0.2.13 eq www
remark adm
permit tcp 10.128.5.0 0.0.0.255 host 192.0.2.14 eq www
remark admin
permit ip host 10.128.6.200 any
```

Рис. 11: Проверка настройки на маршрутизаторе сети «Донская» NAT с правилами

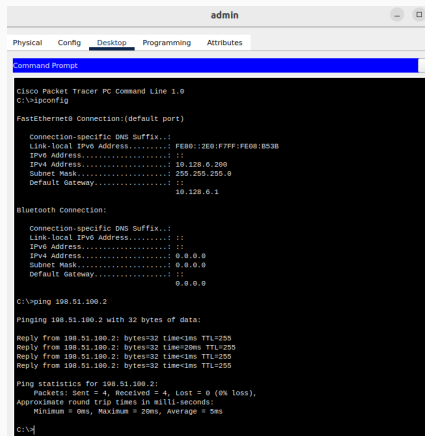
Выполнение лабораторной работы

```
msk-donskaya-onabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-onabakumova-gw-1(config)#ip nat inside source list nat-inet pool main-pool overload
msk-donskaya-onabakumova-gw-1(config)#int f0/0.3
msk-donskaya-onabakumova-gw-1(config-subif)#ip nat inside
msk-donskaya-onabakumova-gw-1(config-subif)#interface f0/0.101
msk-donskaya-onabakumova-gw-1(config-subif)#ip nat inside
msk-donskaya-onabakumova-gw-1(config-subif)#exit
msk-donskaya-onabakumova-gw-1(config)#interface f0/0.102
msk-donskaya-onabakumova-gw-1(config-subif)#ip nat inside
msk-donskaya-onabakumova-gw-1(config-subif)#exit
msk-donskaya-onabakumova-gw-1(config)#interface f0/0.103
msk-donskaya-onabakumova-gw-1(config-subif)#ip nat inside
msk-donskaya-onabakumova-gw-1(config-subif)#exit
msk-donskaya-onabakumova-gw-1(config)#interface f0/0.104
msk-donskaya-onabakumova-gw-1(config-subif)#ip nat inside
msk-donskaya-onabakumova-gw-1(config-subif)#exit
msk-donskaya-onabakumova-gw-1(config)#interface f0/1.4
msk-donskaya-onabakumova-gw-1(config-subif)#ip nat outside
msk-donskaya-onabakumova-gw-1(config-subif)#exit
msk-donskaya-onabakumova-gw-1(config)#^Z
msk-donskaya-onabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-onabakumova-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-onabakumova-gw-1#
```

Рис. 12: Настройка NAT

Выполнение лабораторной работы



```
admin
Physical Config Desktop Programming Attributes
Command Prompt X
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2E0:F7FF:FE08:B53B
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 10.128.0.200
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .:
                                10.128.0.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

C:\>ping 198.51.100.2

Pinging 198.51.100.2 with 32 bytes of data:

Reply from 198.51.100.2: bytes=32 time<1ms TTL=255
Reply from 198.51.100.2: bytes=32 time=20ms TTL=255
Reply from 198.51.100.2: bytes=32 time<1ms TTL=255
Reply from 198.51.100.2: bytes=32 time<1ms TTL=255

Ping statistics for 198.51.100.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 20ms, Average = 5ms

C:\>
```

Рис. 13: Пингование msk-donskaya-omabakumova-gw-1

Выполнение лабораторной работы

```
C:\>ping 198.51.100.1

Pinging 198.51.100.1 with 32 bytes of data:

Reply from 198.51.100.1: bytes=32 time=1ms TTL=254
Reply from 198.51.100.1: bytes=32 time=1ms TTL=254
Reply from 198.51.100.1: bytes=32 time=1ms TTL=254
Reply from 198.51.100.1: bytes=32 time=1ms TTL=254

Ping statistics for 198.51.100.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.0.2.11

Pinging 192.0.2.11 with 32 bytes of data:

Request timed out.
Reply from 192.0.2.11: bytes=32 time=1ms TTL=126
Reply from 192.0.2.11: bytes=32 time=1ms TTL=126
Reply from 192.0.2.11: bytes=32 time=1ms TTL=126

Ping statistics for 192.0.2.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.0.2.11

Pinging 192.0.2.11 with 32 bytes of data:

Reply from 192.0.2.11: bytes=32 time=1ms TTL=126
Reply from 192.0.2.11: bytes=32 time=1ms TTL=126
Reply from 192.0.2.11: bytes=32 time=1ms TTL=126
Reply from 192.0.2.11: bytes=32 time=1ms TTL=126

Ping statistics for 192.0.2.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping www.yandex.ru

Pinging 192.0.2.11 with 32 bytes of data:

Reply from 192.0.2.11: bytes=32 time=10ms TTL=126
Reply from 192.0.2.11: bytes=32 time=1ms TTL=126
Reply from 192.0.2.11: bytes=32 time=1ms TTL=126
Reply from 192.0.2.11: bytes=32 time=1ms TTL=126

Ping statistics for 192.0.2.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>
```

Рис. 14: Пингование маршрутизатора провайдера и сервера www.yandex.ru

Выполнение лабораторной работы

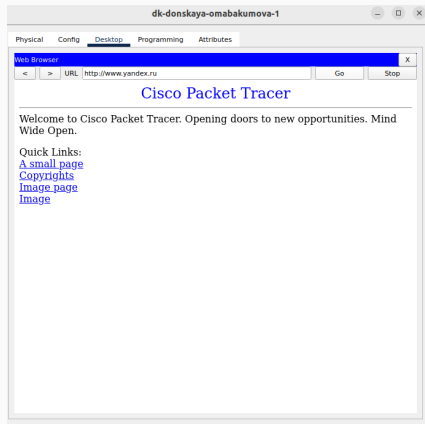


Рис. 15: Запрос в Web-Browser www.yandex.ru

Выполнение лабораторной работы

```
--\nmsk-donskaya-onabakumova-gw-1#conf t\nEnter configuration commands, one per line. End with CNTL/Z.\nmsk-donskaya-onabakumova-gw-1(config)#ip nat inside source static tcp 10.128.0.2 80 198.51.100.2 80\nmsk-donskaya-onabakumova-gw-1(config)#ip nat inside source static tcp 10.128.0.3 20 198.51.100.3 20\nmsk-donskaya-onabakumova-gw-1(config)#ip nat inside source static tcp 10.128.0.3 21 198.51.100.3 21\nmsk-donskaya-onabakumova-gw-1(config)#ip nat inside source static tcp 10.128.0.4 25 198.51.100.4 25\nmsk-donskaya-onabakumova-gw-1(config)#ip nat inside source static tcp 10.128.0.4 110 198.51.100.4 110\nmsk-donskaya-onabakumova-gw-1(config)#ip nat inside source static tcp 10.128.6.200 3389 198.51.100.10 3389\nmsk-donskaya-onabakumova-gw-1(config)#exit\nmsk-donskaya-onabakumova-gw-1#\n%SYS-5-CONF16_I: Configured from console by console\n\nmsk-donskaya-onabakumova-gw-1#wr mem\nBuilding configuration...\n[OK]\nmsk-donskaya-onabakumova-gw-1#
```

Рис. 16: Настройка доступа из Интернета

Выполнение лабораторной работы

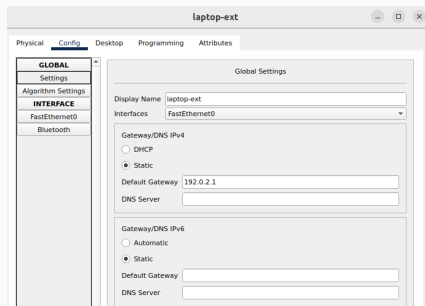


Рис. 17: Задание шлюза на laptop-ext

Выполнение лабораторной работы

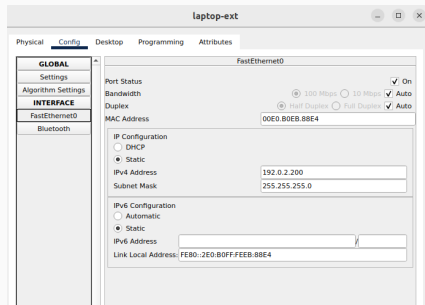


Рис. 18: Задание статического адреса и маски на laptop-ext

Выполнение лабораторной работы

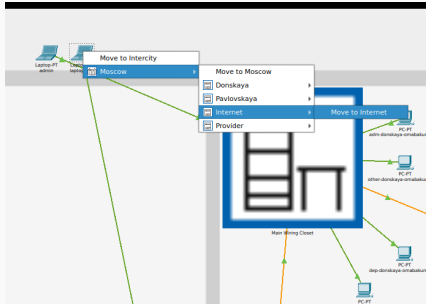
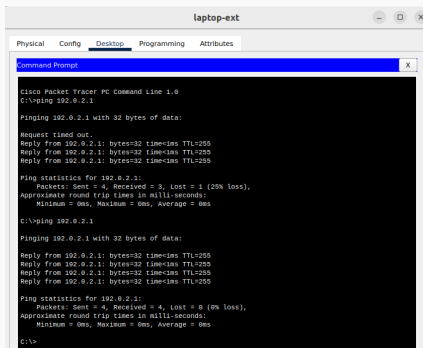


Рис. 19: Перемещение laptop-ext

Выполнение лабораторной работы



The screenshot shows a window titled 'laptop-ext' with tabs for Physical, Config, Desktop, Programming, and Attributes. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the execution of the 'ping' command to 192.0.2.1. The first attempt shows a 'Request timed out.' followed by three successful replies. The second attempt shows four successful replies. The statistics for both attempts indicate 0% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.0.2.1

Pinging 192.0.2.1 with 32 bytes of data:

Request timed out.
Reply from 192.0.2.1: bytes=32 time<1ms TTL=255
Reply from 192.0.2.1: bytes=32 time<1ms TTL=255
Reply from 192.0.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.0.2.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.0.2.1

Pinging 192.0.2.1 with 32 bytes of data:

Reply from 192.0.2.1: bytes=32 time<1ms TTL=255
Reply from 192.0.2.1: bytes=32 time<1ms TTL=255
Reply from 192.0.2.1: bytes=32 time<1ms TTL=255
Reply from 192.0.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.0.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Рис. 20: Пинг проходит успешно

Выполнение лабораторной работы

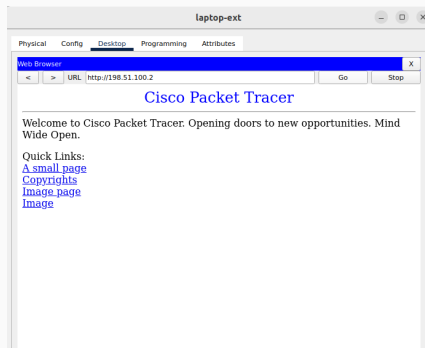
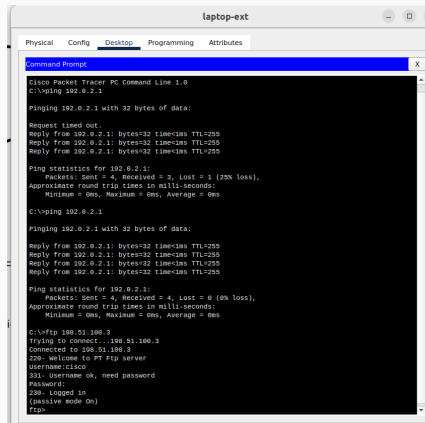


Рис. 21: Страница в Web-Browser открылась успешно

Выполнение лабораторной работы



```
laptop-ext
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.0.2.1

Pinging 192.0.2.1 with 32 bytes of data:

Request timed out.
Reply from 192.0.2.1: bytes=32 time=1ms TTL=255
Reply from 192.0.2.1: bytes=32 time=1ms TTL=255
Reply from 192.0.2.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.0.2.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.0.2.1

Pinging 192.0.2.1 with 32 bytes of data:

Reply from 192.0.2.1: bytes=32 time=1ms TTL=255
Reply from 192.0.2.1: bytes=32 time=1ms TTL=255
Reply from 192.0.2.1: bytes=32 time=1ms TTL=255
Reply from 192.0.2.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.0.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ftp 198.51.100.3
Trying to connect...198.51.100.3
Connected to 198.51.100.3
220- Welcome to PT Ftp server
Username:cisco
331- Username ok, need password
Password:
230- Logged in
(passive mode on)
ftp>
```

Рис. 22: Подключение по FTP прошло успешно

Выводы

Во время выполнения данной лабораторной работы я приобрела практические навыки по настройке доступа локальной сети к внешней сети посредством NAT.